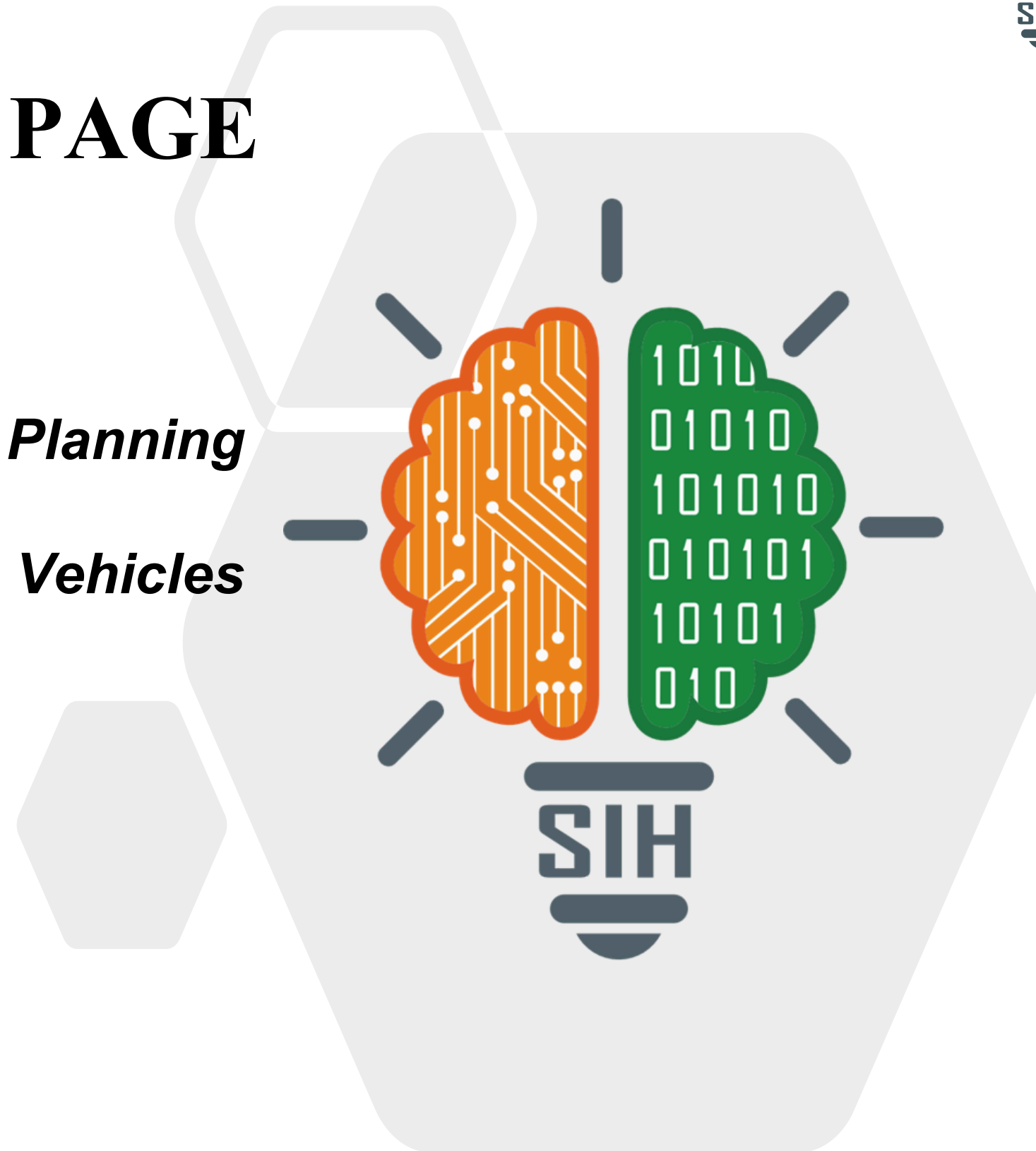


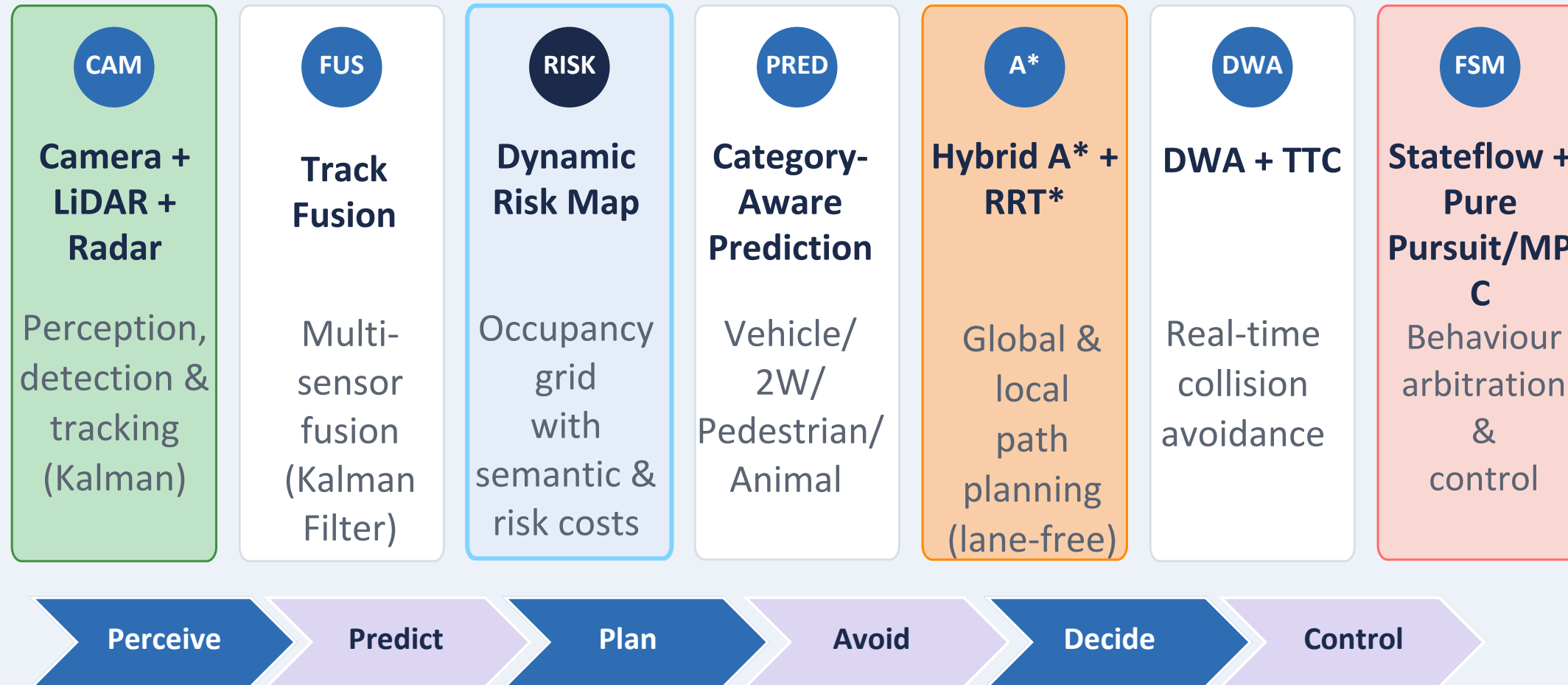
TITLE PAGE

- Problem Statement ID – *PS 26037*
- Problem Statement Title - *Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads*
- Theme- *Smart Vehicles*
- PS Category- *Software*
- Team ID-
- Team Name : *Second Inning*



PROPOSED SOLUTION

Sensor-driven, risk-aware and adaptive planning in MATLAB/Simulink



VALIDATION

Unmarked Village Road

Unsignalized Urban Intersection

Highway Slow-Merge

Dense Market

Sudden Cattle Crossing

✓
Collision-free completion rate

T
Replanning latency < 200 ms (target)

↔
Minimum clearance / TTC

~
Path smoothness (low curvature & jerk)

HOW IT ADDRESSES THE PROBLEM

- **Lane-free planning** : Occupancy-grid based — no lane-graph or HD maps needed (IDD label distributions differ sharply from Cityscapes).
- **Heterogeneous traffic** :Category-aware priors for autos, two-wheelers, carts, pedestrians and animals (TraPHic: up to 30% prediction gain).
- **Road hazards** : Potholes and road edges costed directly into the risk map as planning inputs, not scene decoration.
- **Informal merges & unsignalized junctions** : TTC-triggered yield / slow-down / replan via Stateflow behaviour rules.

INNOVATION & UNIQUENESS

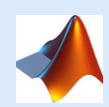
- 01 Integration, not invention**
Established methods (Hybrid A*, RRT*, DWA, MPC) unified and tuned to Indian traffic taxonomies.
- 02 Unified dynamic risk map**
Combines dynamic-agent risk, road hazards, wrong-way risk and prediction uncertainty in one input.
- 03 Category/behaviour-aware prediction**
Agent-specific motion priors using insights from IDD / IDD-3D / TraPHic.
- 04 Baseline isolation + ablation/stress testing**
Quantitative evaluation against named baselines across five Indian scenarios in MATLAB/Simulink.

PERCEIVE → PREDICT → PLAN → DECIDE → ACT → MONITOR → REPLAN

INDIAN ROADS • REAL SCENARIOS • BIGGER IMPACT

TECHNOLOGY STACK

Powered by the MathWorks Ecosystem



MATLAB / Simulink

- End-to-end modeling, simulation
- Algorithm design, testing and validation



RoadRunner

- Creating realistic Indian road scenarios
- Unstructured road, Traffic, Assets, Environment



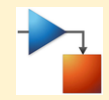
Automated Driving Toolbox

- Camera, LiDAR & Radar sensor simulation
- Object detection, tracking, fusion



Navigation Toolbox

- Hybrid A* & RRT* for robust path planning
- Costmap-based navigation for unstructured roads



Stateflow

- Behavior planning & decision logic
- Safe driving maneuver control



Sensor Fusion & Tracking Toolbox

- Camera–LiDAR–Radar multi-sensor fusion
- Heterogeneous object tracking & state estimation



Computer Vision Toolbox

- Vision-based object detection & road perception
- Pothole, anomaly & road-edge detection

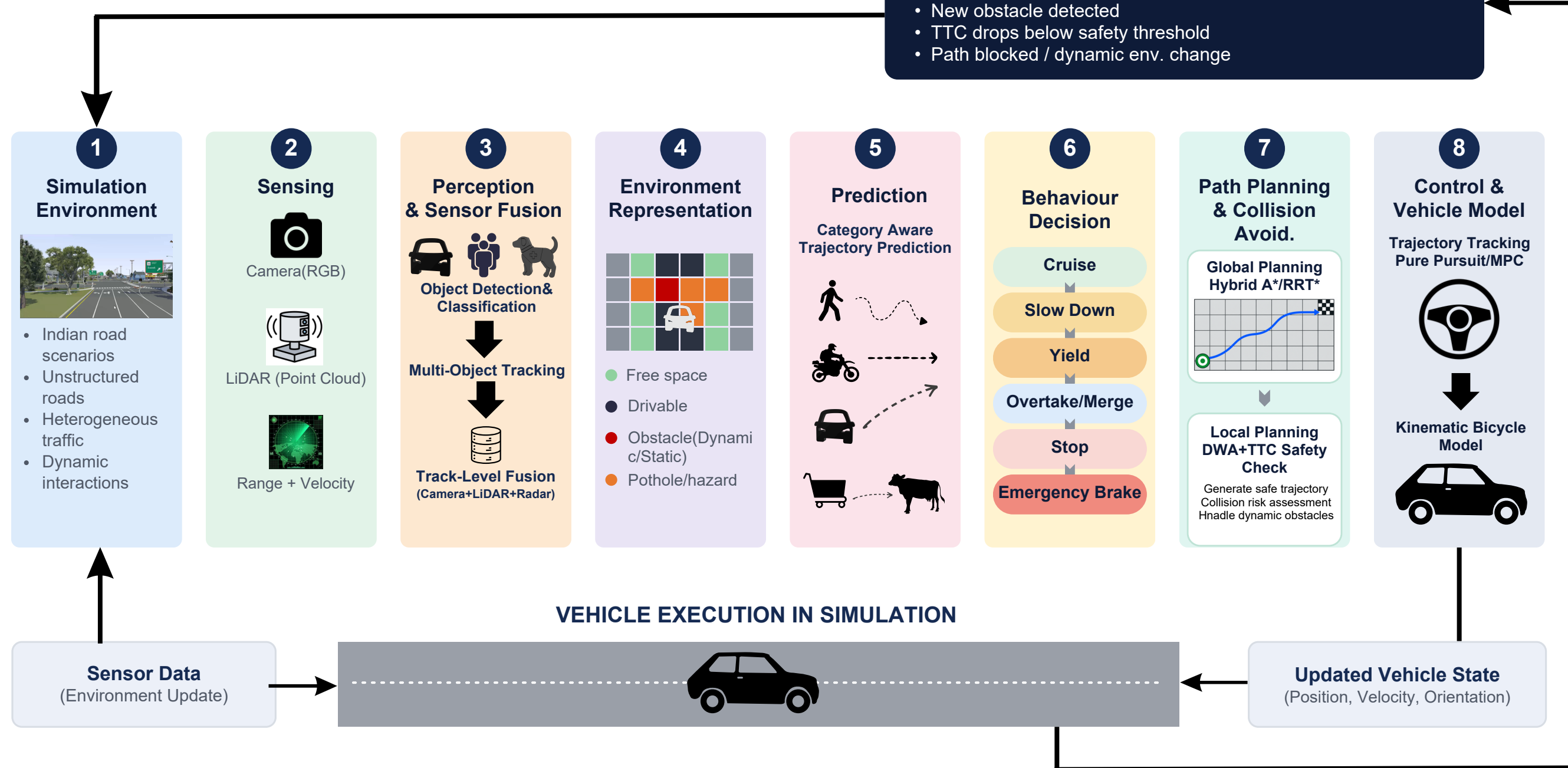


Additional Tools / Models

- Deep Learning Toolbox for prediction
- Vehicle dynamics & simulation analysis

SYSTEM ARCHITECTURE

Perceive → Predict → Plan → Decide → Act — validated across 5 Indian road scenarios



1 The problem is real and measurable India, 2023 · MoRTH data

1,72,890

road deaths in 2023 — highest ever
474 a day · one every 3 minutes

68.5%

of road deaths happen in rural India
unmarked village roads = our Scenario 1

+7.3%

rise in pedestrian deaths in a year
32,825 (2022) → 35,221 (2023)

WHO DIES ON INDIAN ROADS — SHARE OF 2023 DEATHS

Two-wheeler riders 44.8%

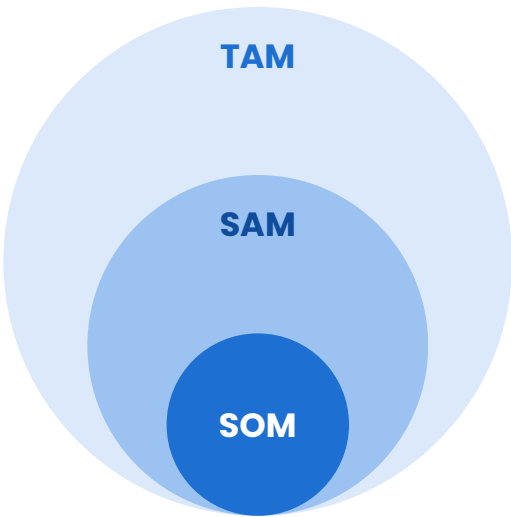
Pedestrians 20.4%

All others 34.8%

≈ 65% are exactly the non-lane, mixed-traffic agents AdaptaDrive is built around

Lane-centric AV stacks (Apollo, Autoware) show disproportionately high lateral collision rates with these road users — HetroD, 2026

2 Market size TAM → SAM → SOM



TAM · GLOBAL ADAS SIMULATION

US\$3.8–3.9 B

in 2025 → US\$9.7–14.3 B by 2032–35 (13–14% CAGR)

SAM · INDIA ADAS / AV

US\$1.15–2.9 B

in 2025 · India AV alone: US\$2.08 B (2023) → 11.37 B (2030)

SOM · FIRST BUYERS

Named, not guessed

TiHAN-type DST hubs (TiHAN, IIT-H: ₹135 Cr), Tier-1s, OEM validation teams

3 Revenue & business model

Who pays? AV testbeds, Tier-1 suppliers and OEM validation teams — for an India-ready scenario library they would otherwise spend months building.

1 B2B licence

Scenario library + risk-map & prediction modules as a MATLAB/Simulink add-on

Licence

2 Govt / research contracts

Pilots with DST NM-ICPS hubs like TiHAN — startup grants up to ₹15 lakh

Open call

3 Scenario subscription

A growing India scenario library for AV labs and OEMs

US\$15k–200k / yr*

Honest timing: revenue follows proven validation — a post-hackathon path, not a Day-1 claim.

4 Scalability MVP → National

National

Shared validation resource for DST hubs & Tier-1 R&D

State

State road types: hill, coastal, metro junctions

Organisation

Partner testbed authors & validates its own scenarios

Pilot

Full N-run validation + one partner AV lab

MVP

36-hour SIH build · all 5 PS scenarios in simulation

New road type or agent class = data cost, not re-engineering

Execution feasibility

Set up

Toolchain, sensors, first Hybrid A* path

Close the loop

DWA + TTC, Stateflow, prediction on 2 scenes

Validate

Scenarios 3–5, multi-run trials, demo video

₹0 hardware — simulation only; licences via SIH (to confirm)

Fallbacks ready — Driving Scenario Designer, Kalman baseline



1,77,175

Road accident deaths
in India (MoRTH 2024)



70.8%

Deaths in rural areas



~ 2 in 3

are Vulnerable Road Users
(Pedestrians, two-wheelers, etc.)

“We’re building the tools to test
autonomy’s safety, scenario by scenario.”



SAFER ROADS

Avoids accidents on
messy, unmarked roads

- Spots and avoids people, animals, and vehicles ahead of time
- Reacts instantly and re-plans its path when something changes

**KPI: % of trips completed
with zero collisions**
(vs. lane-based baseline)

Validated (in simulation)



INCLUSIVE MOBILITY

Works for every kind of
road user

- Understands two-wheelers, pedestrians, carts, and animals — not just cars
- Built for Indian village and semi-urban roads, not just cities

KPI: % of scenarios completed safely
(vs. lane-based baseline)

Validated (in simulation)



SMARTER TRANSPORT

Handles crowded, chaotic
traffic smoothly

- Navigates busy markets and unsignaled junctions safely
- Useful for delivery vehicles and shuttle services

**KPI: Speed of reaction to new
obstacles**
(vs. lane-based baseline)

Validated (in simulation)



EFFICIENCY + SUSTAINABILITY

Smarter driving saves
money and fuel

- Fewer accidents means fewer economic losses
- Smoother driving can mean less fuel wasted (future potential)

KPI: Smoothness of the ride
(fuel/emissions impact: future potential)

Future potential
(to be validated on road)



"A SIMULATION FRAMEWORK TO VALIDATE AV READINESS FOR INDIA"

One reproducible MATLAB/Simulink workflow
for Indian road scenarios

Simulation-validated on MATLAB/Simulink
+ RoadRunner — not field-tested.



5 representative
Indian scenarios



Reusable for research,
OEMs and policy evaluation



Scalable to future
autonomous mobility

- Dolgov, D., Thrun, S., Montemerlo, M., Diebel, J. (2010). Path Planning for Autonomous Vehicles in Unknown Semi-Structured Environments. [CLICK HERE](#)
- Fox, D., Burgard, W., Thrun, S. (1997). The Dynamic Window Approach to Collision Avoidance. IEEE Robotics & Automation Magazine. [CLICK HERE](#)
- Chandra, R. et al. (2019). TraPHic: Trajectory Prediction in Dense and Heterogeneous Traffic Using Weighted Interactions. [CLICK HERE](#)
- Ma, Y. et al. (2019). TrafficPredict: Trajectory Prediction for Heterogeneous Traffic-Agents. [CLICK HERE](#)
- HetroD: A High-Fidelity Drone Dataset and Benchmark for Autonomous Driving in Heterogeneous Traffic (2026). [CLICK HERE](#)
- Fiorini, P., Shiller, Z. (1998). Motion Planning in Dynamic Environments Using Velocity Obstacles. International Journal of Robotics Research. [CLICK HERE](#)